Test

15 August 2025 20:43

Subject: Computer Applications **Max. Marks: 50**
Class: IX **Time: 1.5 hours**

Topics:

1. **Datatypes**
2. **Operators**
3. **Selection**
4. **Loops**

Section A contains 10 questions 2 marks each. You have to attempt all the questions

Section B contains 4 questions 10 marks each, you have to attempt only 3 questions

Section A
[Attempt all questions] *[2x10 =20]*

Q1. Convert the following into a while loop:

```
for (int i=10;i>4;i=i/2)
i=i-1;
System.out.println("Done");
```

Q2. Give and explain the output:

```
char ch='I';
int n=ch;
System.out.print(n+4);
```

Q3. Write a java statement to create an object named "Cricket" of a class "Sports".

Q4. What is an Infinite loop? Give an example of an infinite loop using while loop.

Q5. Write the output of the following code:

```
for (int i=5;i<20;i=i+3)
{
        if (i%2==0)
        System.out.println("J");
        if (i%4==0)
        System.out.println("B");
}
```

Q6. Write Java expression for the following:

$$\sqrt[3]{x^2} - \frac{y}{4z}$$

Q7. Write a statement to print the following including all the quotes and symbols:

He said, " Here's my book/pen ".

Q8. If x=4, y=2 then evaluate the following and find the final value of y:

y += ++x – x++ + --x

Q9. Write the corrected code and underline the corrections:

```
int x=4,y=6;
double z=math.power(x*2,y/3);
```

Q10.Arrange the following operators as per their precedence in ascending order:

( ) , != , % ,&&

Section B

[Attempt any 3 questions] **[3 x 10 = 30]**

The answers in this Section should consist of the Programs in Blue J environment or any program environment with Java as the base.

Q11. A **Tech Number** has even number of digits and if the number is split into two halves, then the square of sum of these halves is equal to the number itself. Write a program in Java to generate and print all four digit tech numbers.

Example:

Square of sum of the halves of 3025 = $(30 + 25)^2$

$=(55)^2$

$=3025$

Q12. Write a menu driven JAVA program to find the sum of the following series till mentioned term:

Choice 1

$$S = \frac{x^3}{15} - \frac{x^6}{11} + \frac{x^9}{7} - \frac{x^{12}}{3} \text{ till 10 terms}$$

Choice 2

$S = 2 + 5 + 10 + 17 + 26 + 37$ .....till T terms (T will be given by user)

Q13. Write a JAVA program to accept annual salary of an employee. The program will calculate and display the income tax paid by the employee based on the following chart:

Annual Salary	**Rate of Income Tax**
Up to 2,50,000	No Tax
2,50,001 to 5,00,000	10% of the amount exceeding 2,50,000
5,00,001 to 10,00,000	Rs 5000 + 20% of the amount exceeding 5,00,000
Above 10,00,000	Rs 10000 + 30% of the amount exceeding 10,00,000

Q14. Pronic number is a number which is the product of two consecutive integers, that is, a number n is a product of x and (x+1). Write a program in Java to check and print whether the number is **Pronic Number** or not.

Example:

Input :56

Output :Pronic Number

Explanation: 56 = 7 * 8 i.e 56 is a product of two consecutive integers 7 and 8.

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